

1. Create a python script file called backups.py (15 points)
 - a. Using a try... except block, attempt to the contents of the file called backups.txt into a variable called data
 - i. If the "FileNotFoundError" error is hit, you should print the following message to the user:
 1. "File not found, please ensure the file exists."
 - ii. Any other error, you should bind the exception message to a variable "e" and print the following message:
 1. "Unknown error occurred: <exception message>"
 - iii. If the try block is successful, do the following:
 1. backups.txt has a list of device names in it with the following format:
(rtr|swt)-(cinoh|daltx)-<5 numbers>-(a|b)
EX:
 - rtr-cinoh-12345-a
 - rtr-daltx-11112-b
 - swt-cinoh-45363-b
 2. Using regular expressions (failure to do so will result in no credit)
 - a. Find all device names that start with rtr and are in cinoh and save to a list called: all_cinoh_rtrs (remember, you want to find the entire name, not just rtr)
 - b. Find all devices that start with swt and are in daltx and save to a list called all_daltx_swt (remember you want to find the entire name, not just daltx)
 - c. Find all devices that that have less than 5 digits in the name and save them to a list called old_names (remember you want to find the entire name, not just the character "a")
 - If you use findall method and captured groups, it will return back a list of tuples where the tuple is a listing of the captured groups from the regex... If this is happening with your code, make sure you make a new list with just the device names and print that.
 3. Print all 3 lists
 - iv. Using a feature of the try block, print the following message regardless of if an exception was raised or not:
 1. "Wrapping up the script. Have a nice day!"
 2. Create a python script file called test_averages.py (15 points)

- a. Create a function called “average” which takes an unlimited number of integers and calculates the average of the numbers and returns that average
 - i. You should use a try...except block to ensure all items passed in are integers
 - ii. if an item passed in is not an integer, you should raise a “TypeError” exception
 - iii. If an exception is raised, you should return “Error”
 - iv. Write 3 test cases (functions):
 1. Function called “test_all_ints”:
 - a. This should test that when you call “average” with 1 and 3, it returns 2.0
 2. Function called “test_floats_and_ints”:
 - a. This should test that when you call average with 1, 4.3, 5.9, and 2, it returns 3.3
 3. Function called “test_bad_input”:
 - a. This should test that when you call “average” with “a” and 3, it returns “Error”